

*Congratulations on being selected for the next round, we are interested in your thought process and thinking when making decisions while building. Answer in the following format: Summary of the work, Implementation and Approach and Upload code to GitHub. If you are selected you will be informed via email and requested to interview at the office. *You will need to bring your laptop.*

Q. Design and implement a Retrieval-Augmented Generation (RAG) pipeline that ingests government PDF documents (both scanned and digitally-born), classifies them by document type, stores them with structured legal metadata, and answers user queries with precise citations including circular/notification numbers, dates, and supersession warnings when a newer document overrides an older one on the same topic.

Core Components

- **Ingestion:** Load texts (e.g., via URLs or PDFs), chunk into 500-1000 token pieces, embed with HuggingFace, store in vector DB (Qdrant) with metadata.
- **Retrieval:** Query vector store for top-k chunks (k=10-20), filter by multiple sources if specified.
- **Augmentation & Comparison:** Prompt LLM with retrieved contexts grouped by source to identify differences (e.g., "Compare facts/events in Source A vs B").
- **Generation:** Output highlighted diffs in markdown.